

56-Mahbub

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2025

Course Title: Database Management System

Course Code: CSE0612215

Marks: 50

Time: 3(three) hours

(Answer all questions)

Q.No.

Questions

Marks

1. Consider the Entity Relationship Diagram shown in Figure 1.

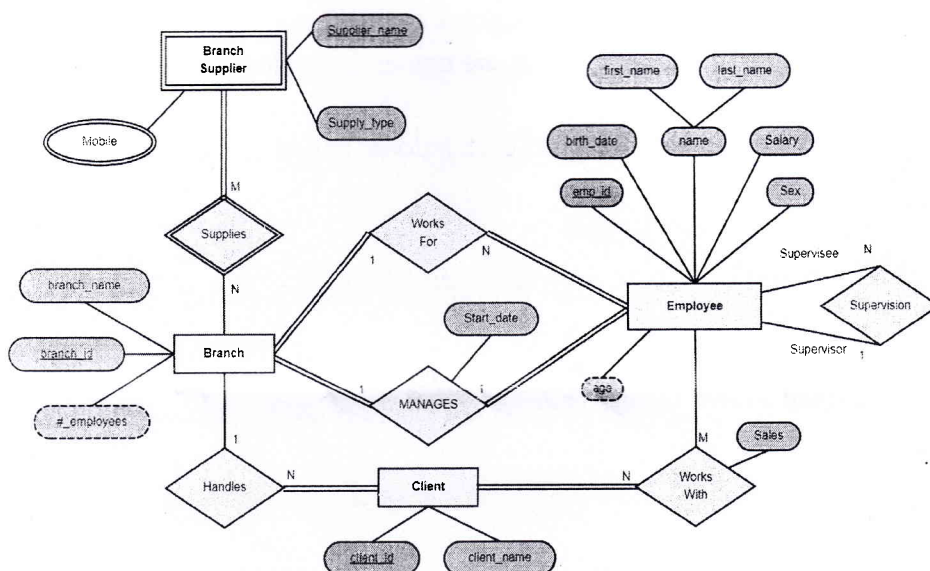


Figure 1

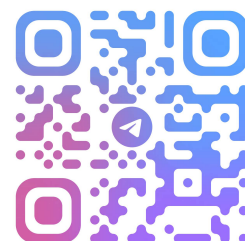
a) Design the Relational Schema from the Entity Relationship Diagram shown in Figure 1. [05]

b) Write the SQL from Figure 1: [05]

- Find the total number of employees.
- List all client names along with the branch name that handles them.

c) Consider the following relational schema and generate the Entity Relationship diagram: [10]

- Customer: customer_id (PK), name, email, phone, address
- Seller: seller_id (PK), name, email, rating
- Product: product_id (PK), name, price, stock_quantity, category
- Order: order_id (PK), customer_id (FK), order_date, total_amount, status
- Payment: payment_id (PK), order_id (FK), payment_date, amount



3. A university database stores the following information:
Student (student_id, student_name, department)
Course (course_id, course_name, credit)
Enrollment (student_id, course_id, marks)

a) Formulate the following SQL from the relational schema:

[04]

- i. Write an SQL query to display the average marks obtained by each student along with their name, but only for those who have enrolled in more than one course.
- ii. Write an SQL statement to insert the following three students into the Student table:

ID: S102, Name: "Arif Hossain", Department: CSE
ID: S103, Name: "Nabila Rahman", Department: EEE
ID: S104, Name: "Tanvir Ahmed", Department: BBA

b) Construct the following relational algebra from the schema:

[06]

- i. List all courses along with the names of students who scored above 90 in those courses.
- ii. Find the average marks of each student across all courses.

4. a) Define RAID and briefly explain the levels of RAID.

[04]

b) Consider the concepts of specialization and generalization, and formulate the relational table shown in Figure 2.

[06]

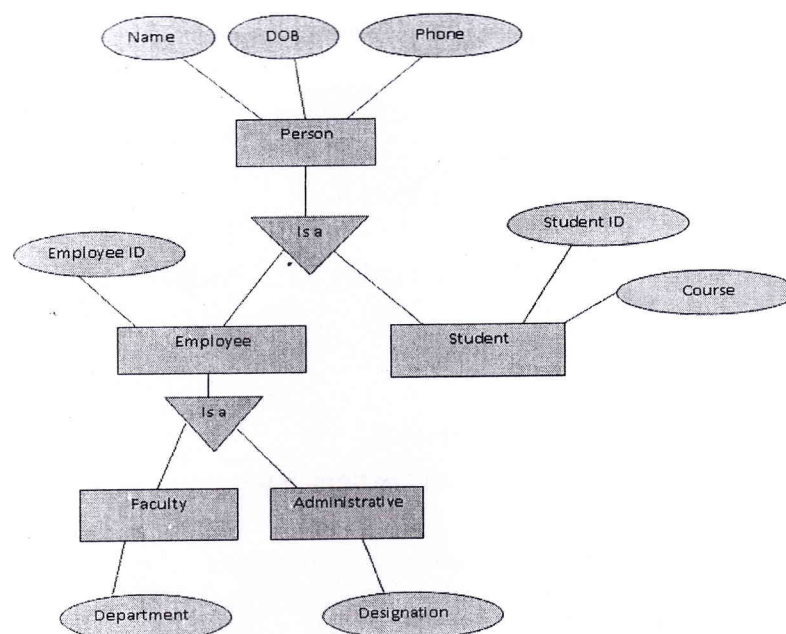
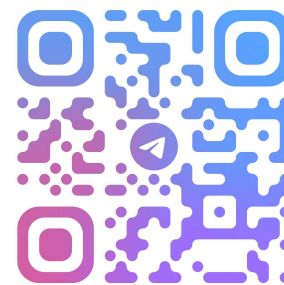


Figure 2



5. Consider the following relational database shown in Table 1.

Table 1

Student_ID	Student_Name	Course_Code	Instructor
S101	Alice	CS101	Dr. Smith
S101	Alice	CS102	Dr. Brown
S102	Bob	CS101	Dr. Smith
S103	Carol	EE101	Dr. Green
S103	Carol	EE102	Dr. White

- a) Define Second Normal Form (2NF) and create a table that satisfies 2NF using Table 1. [04]
- b) Explain why Table 1 violates BCNF and show how it can be decomposed into tables that satisfy BCNF. [06]

